

Solving Systems by Elimination with Scaling

Answer Key

Algebra 1 Linear Systems and Modeling

Quick Answers

1. $x = 3, y = 5$

2. $x = 2, y = 1$

3. $a = 7, c = 5$

4. $x = 2, y = 1$

5. $x = 2, y = -3$

6. $m = 28, s = 22$

7. $x = 3, y = -1$

8. $b = 10, c = 2$

9. $x = 6, y = 2$

10. $x = 5, y = 15$

1. School store: $2x + y = 11, x + 3y = 18$.

Target. The y coefficients are 1 and 3, so scaling the first equation by 3 matches them — cheaper than matching x .

$$\begin{array}{rcl} 3(2x + y) = 3(11) & \implies & 6x + 3y = 33 \\ \text{subtract } x + 3y = 18 & \implies & 5x = 15, \quad x = 3 \end{array}$$

Back-substitute into $2x + y = 11$: $6 + y = 11$, so $y = 5$. Check in the second equation: $3 + 3(5) = 18$ ✓

A notebook costs \$3 and a binder costs \$5.

$$(x, y) = (3, 5)$$

2. $4x - 3y = 5, x + y = 3$.

Target. Scale the second equation by 3 so the y terms are $-3y$ and $+3y$.

$$\begin{array}{rcl} 3(x + y) = 3(3) & \implies & 3x + 3y = 9 \\ \text{add to } 4x - 3y = 5 & \implies & 7x = 14, \quad x = 2 \end{array}$$

Back-substitute into $x + y = 3$: $y = 1$. Check: $4(2) - 3(1) = 5$ ✓

$$(x, y) = (2, 1)$$

3. Theater tickets: $a + c = 12, 9a + 5c = 88$.

Target. Scale the count equation by 5 to match the $5c$ term.

$$\begin{array}{rcl} 5(a + c) = 5(12) & \implies & 5a + 5c = 60 \\ \text{subtract from } 9a + 5c = 88 & \implies & 4a = 28, \quad a = 7 \end{array}$$

Then $c = 12 - 7 = 5$. Check the money equation: $9(7) + 5(5) = 63 + 25 = 88$ ✓

The theater sold 7 adult tickets and 5 child tickets.

$$(a, c) = (7, 5)$$

4. $3x + 4y = 10$, $5x + 6y = 16$.

Target. No coefficient divides another, so scale *both*. For x , the least common multiple of 3 and 5 is 15.

$$\begin{array}{rcl} 5(3x + 4y) = 5(10) & \implies & 15x + 20y = 50 \\ 3(5x + 6y) = 3(16) & \implies & 15x + 18y = 48 \end{array}$$

Subtract: $2y = 2$, so $y = 1$. Back-substitute into $3x + 4y = 10$: $3x = 6$, $x = 2$. Check: $5(2) + 6(1) = 16$ ✓

$$(x, y) = (2, 1)$$

5. $5x + 2y = 4$, $3x + y = 3$.

Target. The y coefficients are 2 and 1, so scale the second equation by 2.

$$\begin{array}{rcl} 2(3x + y) = 2(3) & \implies & 6x + 2y = 6 \\ \text{subtract } 5x + 2y = 4 & \implies & x = 2 \end{array}$$

Back-substitute into $3x + y = 3$: $6 + y = 3$, so $y = -3$. Check the first equation: $5(2) + 2(-3) = 10 - 6 = 4$ ✓

$$(x, y) = (2, -3)$$

6. Bakery: $m + s = 50$, $3m + 4s = 172$.

Target. Scale the count equation by 4 to match the $4s$ term.

$$\begin{array}{rcl} 4(m + s) = 4(50) & \implies & 4m + 4s = 200 \\ \text{subtract from } 3m + 4s = 172 & \implies & -m = -28, \quad m = 28 \end{array}$$

Then $s = 50 - 28 = 22$. Check the money equation: $3(28) + 4(22) = 84 + 88 = 172$ ✓

The bakery sold 28 muffins and 22 scones.

$$(m, s) = (28, 22)$$

7. $4x + 5y = 7$, $6x + 7y = 11$.

Target. Both need scaling. Eliminating x is cheaper: the least common multiple of 4 and 6 is 12, while y would need 35.

$$\begin{array}{rcl} 3(4x + 5y) = 3(7) & \implies & 12x + 15y = 21 \\ 2(6x + 7y) = 2(11) & \implies & 12x + 14y = 22 \end{array}$$

Subtract: $y = -1$. Back-substitute into $4x + 5y = 7$: $4x - 5 = 7$, so $x = 3$. Check: $6(3) + 7(-1) = 18 - 7 = 11$ ✓

$$(x, y) = (3, -1)$$

8. Canoe: $2b + 2c = 24$, $3b - 3c = 24$.

Target. The c coefficients are 2 and -3 ; their least common multiple is 6, so scale both.

$$\begin{array}{rcl} 3(2b + 2c) = 3(24) & \implies & 6b + 6c = 72 \\ 2(3b - 3c) = 2(24) & \implies & 6b - 6c = 48 \end{array}$$

Add: $12b = 120$, so $b = 10$. Back-substitute into $2b + 2c = 24$: $20 + 2c = 24$, so $c = 2$. Check the upstream equation: $3(10) - 3(2) = 24$ ✓

The canoe moves at 10 kilometres per hour in still water and the current runs at 2 kilometres per hour.

$$(b, c) = (10, 2)$$

9. Maya's mistake on $x + 4y = 14$, $3x - 2y = 14$.

(a) What is wrong. Maya multiplied only the left side by 3 and left the right side at 14. That is a different equation, not the same one rewritten: $3x + 12y = 14$ has different solutions from $x + 4y = 14$. Multiplying *every* term — both sides — by 3 is allowed because it scales both sides of a true statement equally, so exactly the same pairs (x, y) still satisfy it. *This part is flagged for manual review; a student answer that says “she has to multiply the 14 too, or the equation is no longer equivalent” earns full credit.*

(b) The correct work.

$$\begin{array}{lll} 3(x + 4y) = 3(14) & \implies & 3x + 12y = 42 \\ \text{subtract } 3x - 2y = 14 & \implies & 14y = 28, \quad y = 2 \end{array}$$

Back-substitute into $x + 4y = 14$: $x + 8 = 14$, so $x = 6$. Check: $3(6) - 2(2) = 18 - 4 = 14$
 $\checkmark$ $(x, y) = (6, 2)$

10. Salt mixture: $x + y = 20$, $30x + 50y = 45 \cdot 20$.

Setting up the second equation. Percent times litres is the salt in each part, and the salt in the two parts must equal the salt in the mixture: $45 \cdot 20 = 900$.

Target. Scale the volume equation by 30 to match the $30x$ term.

$$\begin{array}{lll} 30(x + y) = 30(20) & \implies & 30x + 30y = 600 \\ \text{subtract from } 30x + 50y = 900 & \implies & 20y = 300, \quad y = 15 \end{array}$$

Then $x = 20 - 15 = 5$. Check the salt equation: $30(5) + 50(15) = 150 + 750 = 900$ $\checkmark$
 The technician uses 5 litres of the 30% solution and 15 litres of the 50% solution.

$(x, y) = (5, 15)$